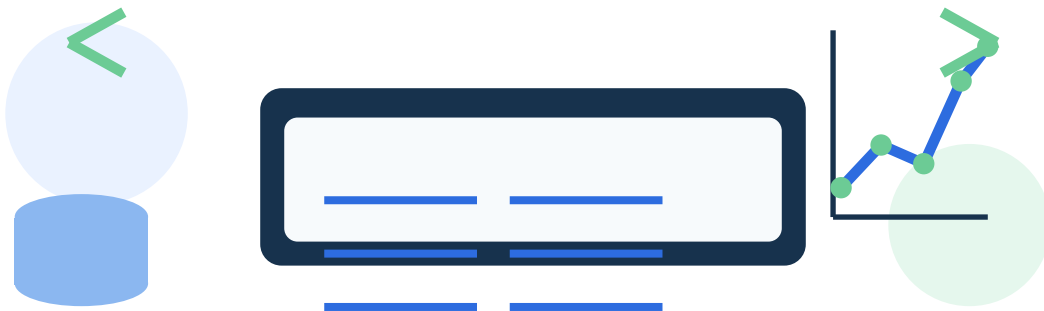

PYTHON FOR DATA SCIENCE & DATA ANALYTICS

87 Multiple Choice Questions



Pandas • NumPy • Data Cleaning • Feature Scaling • Encoding • Core Python • Matplotlib • Seaborn

Answer choices are integrated into each question, with the correct option highlighted for quick revision.

Pandas & NumPy Fundamentals

Q1. What is the primary data structure in Pandas for 2D data?

- A) List
- B) Array
- C) DataFrame**
- D) Series

Pandas & NumPy Fundamentals

Q2. How do you select a single column 'Age' from a DataFrame df?

- A) df['Age'] only
- B) df.Age only
- C) Both A and B**
- D) df.select('Age')

Pandas & NumPy Fundamentals

Q3. Which method is used to read a CSV file in Pandas?

- A) pd.load_csv()
- B) pd.read_csv()**
- C) pd.import_csv()
- D) pd.open_csv()

Pandas & NumPy Fundamentals

Q4. What does `df.head()` return by default?

- A) First 10 rows
- B) First 5 rows**
- C) First row
- D) All rows

Pandas & NumPy Fundamentals

Q5. Which method returns the number of rows and columns in a DataFrame?

- A) `df.dimensions()`
- B) `df.size`
- C) `df.shape`**
- D) `df.count()`

Pandas & NumPy Fundamentals

Q6. How do you rename a column 'old_name' to 'new_name' in a DataFrame?

- A) `df.rename(columns={'old_name': 'new_name'})`**
- B) `df.change_column('old_name', 'new_name')`
- C) `df['old_name'].name = 'new_name'`
- D) `df.columns.rename('old_name', 'new_name')`

Pandas & NumPy Fundamentals

Q7. What is the difference between loc and iloc in Pandas?

- A) No difference
- B) loc uses labels, iloc uses integer positions**
- C) loc is faster than iloc
- D) iloc is deprecated

Pandas & NumPy Fundamentals

Q8. Which method is used to sort a DataFrame by a column 'Age'?

- A) df.sort('Age')
- B) df.order_by('Age')
- C) df.sort_values('Age')**
- D) df.arrange('Age')

Pandas & NumPy Fundamentals

Q9. How do you get unique values in a column 'City'?

- A) df['City'].unique()**
- B) df['City'].distinct()
- C) df['City'].get_unique()
- D) unique(df['City'])

Pandas & NumPy Fundamentals

Q10. What does `df.describe()` return?

- A) First few rows
- B) Data types of columns
- C) Statistical summary of numerical columns**
- D) Column names

Pandas & NumPy Fundamentals

Q11. What is the default data type of a NumPy array created from integers?

- A) float64
- B) int32 or int64 (platform dependent)**
- C) object
- D) string

Data Cleaning & Missing Values

Q12. How do you create a 1D NumPy array with values from 0 to 9?

- A) `np.array(range(10))`
- B) `np.arange(10)`
- C) Both A and B**
- D) `np.range(10)`

Data Cleaning & Missing Values

Q13. What is the shape of `np.zeros((3, 4))`?

A) (3, 4)

B) (4, 3)

C) 12

D) (3,)

Data Cleaning & Missing Values

Q14. Which operation is element-wise in NumPy?

A) `arr1 + arr2`

B) `arr1 * arr2`

C) `arr1 / arr2`

D) All of the above

Data Cleaning & Missing Values

Q15. How do you reshape a 1D array of 12 elements to a 3x4 matrix?

A) `arr.reshape(3, 4)`

B) `arr.shape(3, 4)`

C) `arr.resize(3, 4)`

D) `arr.transform(3, 4)`

Data Cleaning & Missing Values

Q16. What does `np.linspace(0, 10, 5)` create?

A) [0, 2, 4, 6, 8]

B) [0, 2.5, 5, 7.5, 10]

C) [0, 1, 2, 3, 4]

D) [0, 10, 20, 30, 40]

Data Cleaning & Missing Values

Q17. Which method is used to find the maximum value in a NumPy array?

A) `arr.max()`

B) `np.max(arr)`

C) Both A and B

D) `arr.maximum()`

Data Cleaning & Missing Values

Q18. What is broadcasting in NumPy?

A) Sending arrays over network

B) Automatic expansion of arrays for element-wise operations

C) Converting arrays to lists

D) Parallel processing

Data Cleaning & Missing Values

Q19. Which method removes duplicate rows from a DataFrame?

A) `df.remove_duplicates()`

B) `df.drop_duplicates()`

C) `df.unique()`

D) `df.deduplicate()`

Data Cleaning & Missing Values

Q20. How do you check for missing values in a DataFrame?

A) `df.isnull()`

B) `df.isna()`

C) Both A and B

D) `df.missing()`

Data Cleaning & Missing Values

Q21. What does `df.dropna()` do by default?

A) Drops rows with any missing values

B) Drops columns with any missing values

C) Fills missing values with 0

D) Replaces missing values with mean

Data Cleaning & Missing Values

Q22. Which parameter in `dropna()` specifies the minimum number of non-NA values required?

A) minimum

B) thresh

C) limit

D) count

Data Cleaning & Missing Values

Q23. How do you replace all occurrences of 'Unknown' with NaN in a DataFrame?

A) `df.replace('Unknown', np.nan)`

B) `df.substitute('Unknown', np.nan)`

C) `df.change('Unknown', np.nan)`

D) `df.convert('Unknown', np.nan)`

Data Cleaning & Missing Values

Q24. What is the purpose of `df.astype()` method?

A) To sort data

B) To convert data types of columns

C) To filter data

D) To merge dataframes

Data Cleaning & Missing Values

Q25. How do you remove leading and trailing whitespaces from string columns?

A) `df['column'].strip()`

B) `df['column'].str.strip()`

C) `df.strip('column')`

D) `strip(df['column'])`

Data Cleaning & Missing Values

Q26. Which method fills missing values with a specified value?

A) `df.fill()`

B) `df.fillna()`

C) `df.replace_na()`

D) `df.impute()`

Data Cleaning & Missing Values

Q27. How do you fill missing values with the mean of the column?

A) `df.fillna(df.mean())`

B) `df.fillna(mean())`

C) `df.fill_mean()`

D) `df.impute_mean()`

Data Cleaning & Missing Values

Q28. What does the method parameter in fillna() do?

- A) Specifies the filling value
- B) Specifies forward fill or backward fill**
- C) Specifies the column to fill
- D) Specifies the data type

Data Cleaning & Missing Values

Q29. Which value does 'ffill' represent in fillna(method='ffill')?

- A) Fill with first value
- B) Forward fill (propagate last valid observation forward)**
- C) Fill with mean
- D) Fill with zero

Data Cleaning & Missing Values

Q30. How do you interpolate missing values in a time series?

- A) df.interpolate()**
- B) df.fillna(interpolate=True)
- C) df.impute()
- D) df.smooth()

Data Cleaning & Missing Values

Q31. What does `df.dropna(axis=1)` do?

A) Drops rows with missing values

B) Drops columns with missing values

C) Fills missing values

D) Counts missing values

Data Cleaning & Missing Values

Q32. How do you fill missing values differently for each column?

A) Pass a dictionary to `fillna()`

B) Use a loop

C) Both A and B

D) Not possible

Feature Scaling

Q33. What is the purpose of feature scaling?

A) To remove outliers

B) To bring features to similar scales

C) To reduce dimensionality

D) To encode categorical variables

Feature Scaling

Q34. Which scaling method transforms features to a range between 0 and 1?

A) StandardScaler

B) MinMaxScaler

C) RobustScaler

D) Normalizer

Feature Scaling

Q35. What does StandardScaler do?

A) Scales to [0, 1] range

B) Scales to have mean=0 and std=1

C) Scales to [-1, 1] range

D) Removes outliers

Feature Scaling

Q36. Which scaler is most robust to outliers?

A) MinMaxScaler

B) StandardScaler

C) RobustScaler

D) MaxAbsScaler

Feature Scaling

Q37. From which library do you import StandardScaler in scikit-learn?

A) sklearn.preprocessing

B) sklearn.scaling

C) sklearn.transform

D) sklearn.normalize

Feature Scaling

Q38. What is normalization in the context of feature scaling?

A) Same as standardization

B) Scaling individual samples to have unit norm

C) Removing duplicates

D) Handling missing values

Feature Scaling

Q39. When should you use MinMaxScaler?

A) When you want features in a bounded interval

B) When data has outliers

C) When data is normally distributed

D) Never

Feature Scaling

Q40. What happens if you don't scale features before applying KNN or SVM?

A) Nothing, works the same

B) Features with larger scales dominate the distance calculations

C) Algorithm won't run

D) Improves performance

Categorical Encoding

Q41. What is Label Encoding?

A) Encoding numerical values

B) Converting categorical labels to numerical values

C) Creating new features

D) Removing labels

Categorical Encoding

Q42. Which sklearn class is used for Label Encoding?

A) LabelEncoder

B) OneHotEncoder

C) OrdinalEncoder

D) CategoricalEncoder

Categorical Encoding

Q43. What is the main disadvantage of Label Encoding for nominal categorical variables?

- A) Slow performance
- B) Creates ordinal relationship that doesn't exist**
- C) Cannot handle missing values
- D) Uses too much memory

Categorical Encoding

Q44. What does One-Hot Encoding do?

- A) Assigns single number to each category
- B) Creates binary columns for each category**
- C) Removes categorical variables
- D) Sorts categories

Categorical Encoding

Q45. If a categorical feature has 5 unique values, how many columns will One-Hot Encoding create?

- A) 4
- B) 5**
- C) 6
- D) 1

Note: Note: With drop_first=True, it would create 4 columns

Categorical Encoding

Q46. Which pandas method is commonly used for One- Hot Encoding?

A) `pd.encode()`

B) `pd.get_dummies()`

C) `pd.one_hot()`

D) `pd.categorize()`

Categorical Encoding

Q47. What is the "dummy variable trap"?

A) Too many categories

B) Perfect multicollinearity from including all dummy variables

C) Missing values in encoding

D) Encoding errors

Categorical Encoding

Q48. When should you use Label Encoding over One- Hot Encoding?

A) For nominal variables

B) For ordinal variables with inherent order

C) For all categorical variables

D) Never

Pandas Operations & Analysis

Q49. How do you merge two DataFrames df1 and df2 on a common column 'ID'?

A) `pd.merge(df1, df2, on='ID')`

B) `df1.merge(df2, on='ID')`

C) Both A and B

D) `df1.join(df2, key='ID')`

Pandas Operations & Analysis

Q50. What is the difference between `merge()` and `concat()`?

A) No difference

B) `merge()` combines on keys, `concat()` stacks along an axis

C) `concat()` is faster

D) `merge()` only works with 2 DataFrames

Pandas Operations & Analysis

Q51. How do you group data by a column and calculate mean?

A) `df.groupby('column').mean()`

B) `df.group('column').mean()`

C) `df.aggregate('column', 'mean')`

D) `df.summarize('column', 'mean')`

Pandas Operations & Analysis

Q52. What does `df.pivot_table()` do?

A) Rotates the DataFrame

B) Creates a spreadsheet-style pivot table

C) Sorts the data

D) Filters the data

Pandas Operations & Analysis

Q53. How do you apply a custom function to each element of a DataFrame column?

A) `df['column'].map(function)`

B) `df['column'].apply(function)`

C) Both A and B

D) `df['column'].transform(function)`

Pandas Operations & Analysis

Q54. What is the purpose of the `axis` parameter in pandas operations?

A) To specify rotation

B) To specify whether to operate row-wise (0) or column-wise (1)

C) To specify data type

D) To specify sorting order

Pandas Operations & Analysis

Q55. How do you convert a pandas Series to a NumPy array?

- A) `series.to_array()`
- B) `series.values`
- C) `np.array(series)`

D) Both B and C

Pandas Operations & Analysis

Q56. What does `pd.crosstab()` compute?

- A) Cross-validation scores

B) Cross-tabulation of two or more factors

- C) Correlation matrix
- D) Covariance matrix

Pandas Operations & Analysis

Q57. How do you save a DataFrame to a CSV file?

- A) `df.save_csv('file.csv')`

B) `df.to_csv('file.csv')`

- C) `df.write_csv('file.csv')`
- D) `df.export_csv('file.csv')`

Pandas Operations & Analysis

Q58. What is vectorization in pandas/numpy?

- A) Converting to vectors
- B) Performing operations on entire arrays without explicit loops**
- C) 3D operations
- D) Parallel processing

Pandas Operations & Analysis

Q59. How do you filter rows where Age > 25 in a DataFrame?

- A) `df[df['Age'] > 25]`**
- B) `df.filter(Age > 25)`
- C) `df.select(Age > 25)`
- D) `df.where('Age' > 25)`

Pandas Operations & Analysis

Q60. What does `df.corr()` calculate?

- A) Covariance matrix
- B) Correlation matrix**
- C) Confusion matrix
- D) Distance matrix

Pandas Operations & Analysis

Q61. What is the output of `[1, 2, 3] + [4, 5]`?

A) `[1, 2, 3, 4, 5]`

B) `[5, 7, 3]`

C) Error

D) `[[1, 2, 3], [4, 5]]`

Core Python

Q62. What does `list.pop()` do without any argument?

A) Removes first element

B) Removes last element and returns it

C) Removes all elements

D) Returns the list

Core Python

Q63. What is the output of `[1, 2, 3, 4, 5][1:4]`?

A) `[1, 2, 3, 4]`

B) `[2, 3, 4]`

C) `[2, 3, 4, 5]`

D) `[1, 2, 3]`

Core Python

Q64. Which method adds an element to the end of a list?

- A) list.add()
- B) list.insert()
- C) list.append()**
- D) list.push()

Core Python

Q65. What is the time complexity of accessing an element by index in a Python list?

- A) $O(n)$
- B) $O(\log n)$
- C) $O(1)$**
- D) $O(n^2)$

Core Python

Q66. What is the output of $\{1, 2, 3\}$ & $\{2, 3, 4\}$?

- A) $\{1, 2, 3, 4\}$
- B) $\{2, 3\}$**
- C) $\{1, 4\}$
- D) Error

Note: Intersection operation

Core Python

Q67. Which characteristic is TRUE about Python sets?

- A) Ordered collection
- B) Allows duplicate elements
- C) Mutable (can add/remove elements)**
- D) Supports indexing

Core Python

Q68. What does `set1.difference(set2)` return?

- A) Elements in both sets
- B) Elements in set1 but not in set2**
- C) Elements in set2 but not in set1
- D) All elements from both sets

Core Python

Q69. Which operator performs union of two sets?

- A) `&`
- B) `|`**
- C) `^`
- D) `-`

Core Python

Q70. What is the output of `len({1, 2, 2, 3, 3, 3})`?

A) 6

B) 3

C) 4

D) Error

Note: Sets don't allow duplicates

Core Python

Q71. What is the output of `{'a': 1, 'b': 2}.get('c', 0)`?

A) None

B) 0

C) Error

D) 'c'

Core Python

Q72. Which method returns all keys in a dictionary?

A) `dict.keys()`

B) `dict.getkeys()`

C) `dict.all_keys()`

D) `dict.keylist()`

Core Python

Q73. What happens when you access a non-existent key using dict['key']?

A) Returns None

B) Returns 0

C) Raises KeyError

D) Returns empty string

Core Python

Q74. How do you merge two dictionaries dict1 and dict2 (Python 3.9+)?

A) dict1 + dict2

B) dict1 | dict2

C) dict1.merge(dict2)

D) merge(dict1, dict2)

Core Python

Q75. What is the time complexity of dictionary lookup by key?

A) $O(n)$

B) $O(\log n)$

C) $O(1)$ average case

D) $O(n^2)$

Core Python

**Q76. What does the following code output? `def func(x=[]): x.append(1)
return x print(func()) print(func())`**

A) [1] and [1]

B) [1] and [1, 1]

C) Error

D) [] and [1]

Note: Mutable default argument trap

Core Python

Q77. Which approach is most efficient to check if an element exists in a collection?

A) Using a list with in operator

B) Using a set with in operator

C) Using a dictionary with in operator

D) Both B and C

Note: Both are $O(1)$ average case vs $O(n)$ for list

Core Python

**Q78. What is the output of the following? `x = [1, 2, 3] y = x y.append(4)
print(x)`**

A) [1, 2, 3]

B) [1, 2, 3, 4]

C) [4]

D) Error

Note: y is a reference to x, not a copy

Core Python

Q79. How do you find the most frequent element in a list efficiently?

- A) Use max() with count()
- B) Use collections.Counter**
- C) Sort and iterate
- D) Use a dictionary manually

Note: Most Pythonic and efficient

Core Python

Q80. What does this list comprehension do: `[x**2 for x in range(10) if x % 2 == 0]`?

- A) Squares of all numbers from 0 to 9
- B) Squares of even numbers from 0 to 8**
- C) Even numbers from 0 to 9
- D) Squares of odd numbers

Data Visualization & Python Analytics

Q81. Which function is used to create a figure and subplots in Matplotlib?

- A) plt.figure()
- B) plt.subplots()**
- C) plt.create_plot()
- D) plt.subplot()

Note: plt.subplots() returns both figure and axes objects

Data Visualization & Python Analytics

Q82. What is the main advantage of Seaborn over Matplotlib?

- A) Faster performance
- B) Better default aesthetics and statistical plotting capabilities**
- C) More plot types
- D) Works without Matplotlib

Data Visualization & Python Analytics

Q83. Which Seaborn function is best for visualizing the distribution of a numerical variable?

- A) `sns.barplot()`
- B) `sns.scatterplot()`
- C) `sns.histplot()` or `sns.kdeplot()`**
- D) `sns.lineplot()`

Data Visualization & Python Analytics

Q84. How do you create a correlation heatmap in Seaborn?

- A) `sns.heatmap(df.corr())`**
- B) `sns.corrplot(df)`
- C) `sns.correlation(df)`
- D) `plt.heatmap(df)`

Data Visualization & Python Analytics

Q85. Which parameter in Seaborn's pairplot() is used to color points by a categorical variable?

- A) color
- B) palette
- C) hue**
- D) category

Data Visualization & Python Analytics

Q86. What is the output of the following code? import pandas as pd df = pd.DataFrame({'A': [1, 2, 3], 'B': [4, 5, 6]}) print(df.loc[0, 'A'])

- A) 0
- B) 1**
- C) [1, 4]
- D) Error

Note: loc[0, 'A'] accesses row with index 0, column 'A', which is 1

Data Visualization & Python Analytics

Q87. In NumPy, what does the axis parameter represent in np.sum(arr, axis=0)?

- A) Sum along rows (column-wise sum)**
- B) Sum along columns (row-wise sum)
- C) Sum of all elements
- D) Sum of diagonal elements

Note: axis=0 means operation along the first dimension, collapsing rows, resulting in column-wise sum